

HAT-P-54b

R-1700

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_180421 · created 2026-08-02T14:05:39+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458229.62833 +/- 0.00055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.199 +/- 0.054
Transit depth (Rp/Rs)^2	3.95 +/- 2.13 [%]
Orbital Inclination (inc)	85.6 +/- 2.06
Airmass coefficient 1 (a1)	1.339 +/- 0.037
Airmass coefficient 2 (a2)	-0.164 +/- 0.027
Scatter in the residuals of the lightcurve fit is	2.73 %
Best Comparison Star	None
Optimal Aperture	2.81
Optimal Annulus	14.93
Transit Duration (day)	0.0213 +/- 0.0075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.199 ± 0.054 vs 0.1572 (deviation 0.77σ)
Duration fit vs published	0.0213 d vs 0.075 d (deviation 7.16σ)
Mid-time O-C	-2.8 min
Depth SNR	3.7
Residual scatter	2.73 %

Light curve

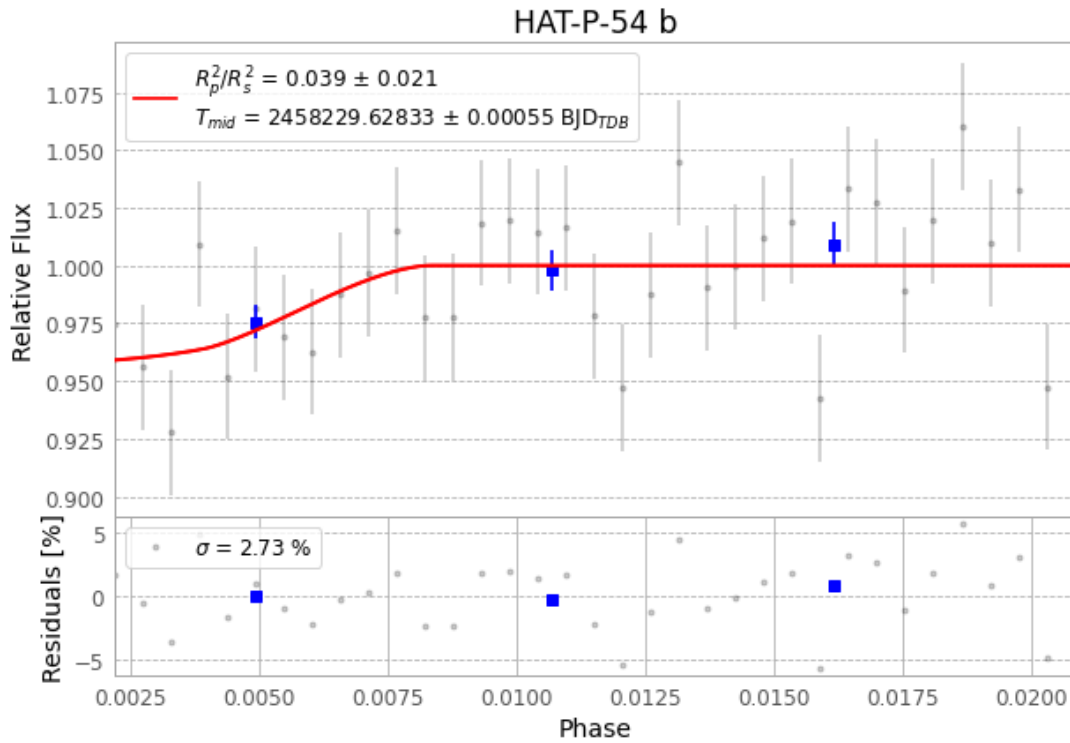


Figure 1 — FinalLightCurve_HAT-P-54 b_2018-04-20.png (SHA-256 a8598a1b247e7ddd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [217, 295], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1232a6bdd369eae5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

35 FITS frames (23 MB), HATP-54180421031812.FITS → HATP-54180421050012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-180421.log (c24994cb2c51...), FinalLightCurve_HAT-P-54 b_2018-04-20.png (a8598a1b247e...), FinalLightCurve_HAT-P-54 b_2018-04-20.csv (0c63ad4081f0...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 14:05Z.